Keshav Sharma

India | 8168289407 | keshavsharma71982@gmail.com | LinkedIn Link | GitHub Link | Portfolio

EDUCATION

Vellore Institute of Technology — **Chennai, Tamil Nadu**

Bachelor of Technology in Computer Science (Data Science Specialisation) — *Graduation: May 2027*

SKILL SUMMARY

Programming Languages: Python, Java, SQL, R, C/C++
Data Science & Machine Learning: Pandas, NumPy, EDA, Feature Engineering, Predictive Modeling, Anomaly Detection
Generative AI & LLMs: Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG)
Databases & Data Engineering: MySQL, SQLite, MongoDB, Redis, SQLAlchemy, Relational Database Design, Data Modeling, Query Optimization, ETL Pipelines
Backend & API Development: FastAPI, Flask, Node.js, Express.js, REST APIs, API Design, Middleware, Authentication, JSON, TCP Sockets
Computer Science Fundamentals: Data Structures and Algorithms, Object-Oriented Programming (OOP), Operating Systems, Computer Networks, Database Management Systems (DBMS), Multithreading, Concurrency, Caching, Exception Handling
Data Visualization & Analytics: Power BI, DAX, Plotly, Dashboard Development, Business Analytics

EXPERIENCE

AI for Sustainability Virtual Intern **|LINK** — ***July – Sep, 2026***

- Completed an AI-focused virtual internship covering Generative AI, Prompt Engineering, RAG, Agentic AI, and Responsible AI, with hands-on exposure to enterprise AI workflows and LLM applications.
- Applied RAG and semantic search concepts using document ingestion, embeddings, vector databases, and retrieval pipelines to improve the grounding and reliability of LLM responses.
- Built an Energy Anomaly Detector using Python, Pandas, NumPy, and Isolation Forest, detecting 9 anomalies across 6,918 test hours (0.13%), with automated email alerts, privacy safeguards, transparent scoring, and rate-limited notifications.

PROJECTS

Financial Transaction Anomaly Detection & Audit Dashboard **| LINK**

- Engineered an end-to-end Python analytics pipeline processing 50K+ PaySim transactions, incorporating data validation, duplicate detection, statistical outlier analysis, vendor-velocity checks, HHI concentration analysis, and Isolation Forest.
- Designed a star-schema SQLite data warehouse and implemented a composite 0–100 risk-scoring framework, identifying 357 high/critical-risk transactions with end-to-end analysis completed in under 7 seconds.
- Developed a 5-page Power BI dashboard with 20+ DAX measures to deliver vendor risk, anomaly-rate, and transaction trend insights, supported by automated pipeline tests.

Docwise **|LINK**

- Built an AI-powered PDF question-answering system using Google Gemini and RAG, enabling contextual question answering over documents exceeding 200 pages.
- Developed a document processing and retrieval pipeline combining PDF extraction, chunking, embeddings, and semantic retrieval, achieving 95%+ query accuracy.
- Integrated FastAPI and LLM inference to deliver an end-to-end AI application for document understanding and conversational querying.

Customer Lifetime Value Engine **|LINK**

- Built BG/NBD and Gamma-Gamma probabilistic models on 1M+ transaction records to predict customer lifetime value and purchasing behaviour, incorporating frequency/recency analysis and model validation.
- Performed transaction-level preprocessing and value-based customer segmentation, identifying the top 20% of customers contributing ~65% of forecasted revenue.
- Developed Power BI dashboards to visualise customer segments, revenue distribution, and lifetime value predictions, enabling data-driven business decisions and customer prioritisation.

RedisLite **|LINK**

- Engineered a Redis-inspired in-memory key-value database in Python featuring a custom RESP protocol parser, multithreaded TCP server, thread-safe storage engine, and an O(1) LRU cache with lazy/active TTL expiration.
- Designed a four-layer architecture (TCP transport, storage engine, cache/TTL, persistence) implementing Write-Ahead Logging (WAL), atomic log compaction, and automatic crash recovery.
- Validated the system with 46 automated PyTest unit tests (TDD), benchmarking 619K SET ops/sec in-memory and 2.8K ops/sec with durable WAL persistence.

Certifications

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional **|LINK** — 2026